# Report on Life Expectancy at Birth (2000–2021)

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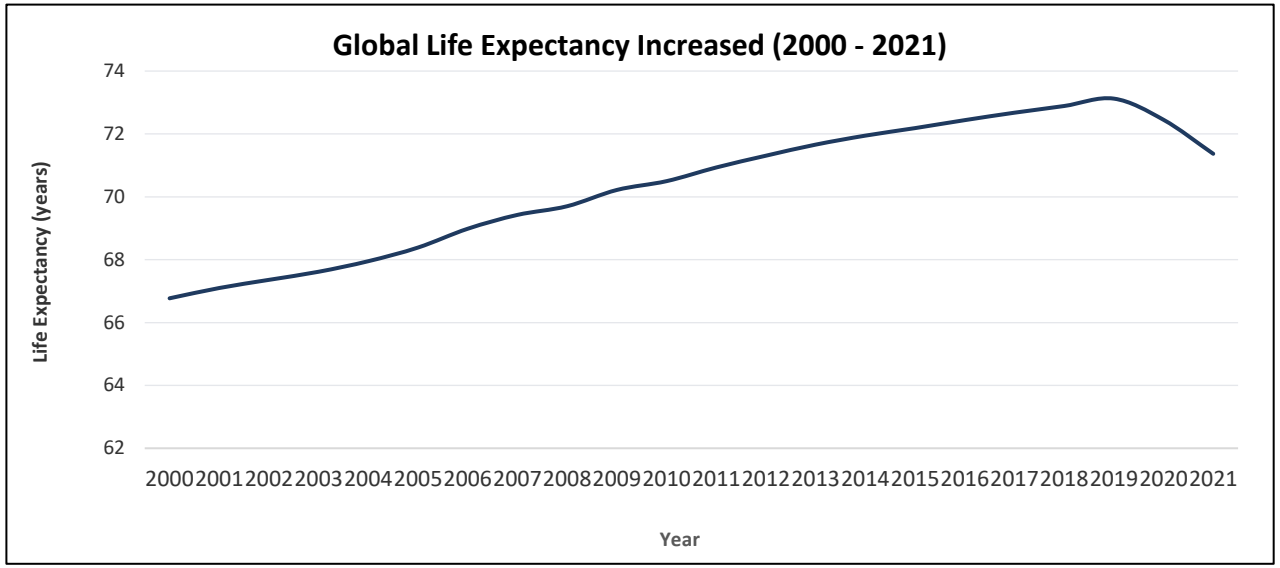
# 1. Introduction

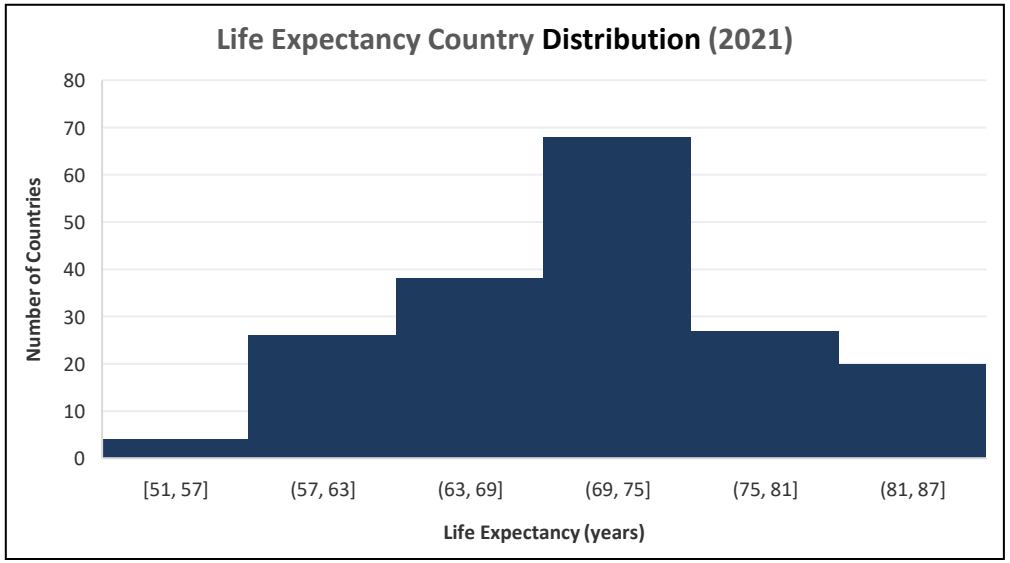
- **Dataset Overview**: The dataset contains life expectancy estimates at birth across countries, regions, and globally, disaggregated by sex (Male, Female, Total) for the years 2000–2021.
- **Purpose of Analysis**: To explore trends, disparities, and patterns in life expectancy using clear visualizations.
- **First Impressions**:
  - Global life expectancy appears to have steadily increased.
  - Significant variation exists between regions and countries.
  - Females generally live longer than males.

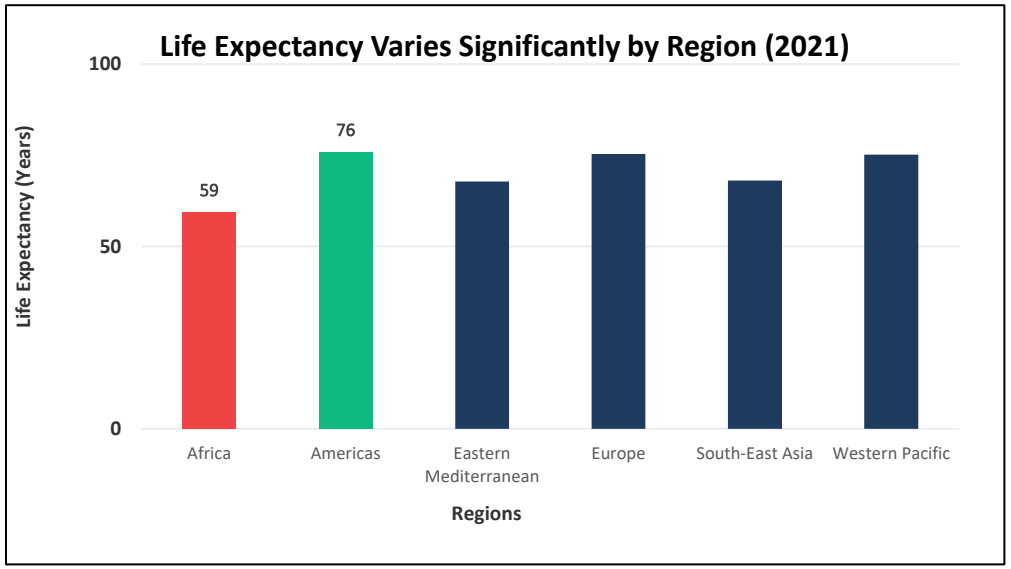
# 2. Questions Explored

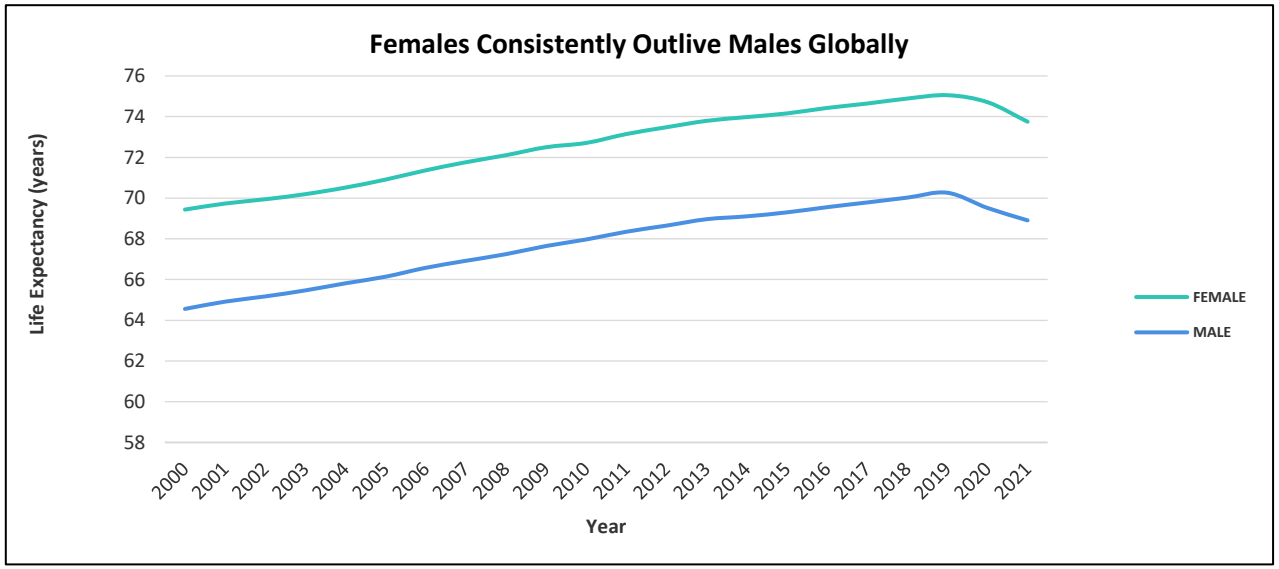
- **Global Trend**: How has life expectancy changed worldwide from 2000–2021?
- **Regional Comparison**: How do WHO regions compare in average life expectancy in 2021?
- **Gender Gap**: What differences exist between male and female life expectancy in selected countries?
- **Country Extremes**: Which countries have the highest and lowest life expectancy in 2021?

# 3. Visualizations



# 4. Key Findings

- **Global Trend**: Life expectancy has risen steadily, reflecting improvements in healthcare and living conditions.
- **Regional Comparison**: Europe and Western Pacific lead with the highest averages, while Africa lags behind.
- **Gender Gap**: Females consistently outlive males by 5–7 years in most countries.
- **Country Extremes**: Japan, Switzerland, and Australia top the list (>82 years), while countries like Chad and Nigeria remain below 60 years.

# 5. Limitations

- Dataset covers only 2000–2021 (no earlier history).
- Regional averages may mask internal disparities within countries.
- Life expectancy does not capture quality of life or inequality within populations.
- No causal variables (e.g., income, healthcare access) included.
- Country-level variation may not be fully represented.
- Confidence intervals (AMOUNT_NL, AMOUNT_NU) were not fully analyzed.

# 6. Conclusion

- This dataset highlights both progress and persistent inequality in global health. While life expectancy has improved worldwide, regional and gender disparities remain significant. These insights can guide further exploration into the social, economic, and healthcare factors driving these differences.
- This also highlights the importance of targeted health interventions and policy improvements.